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RESEARCH ARTICLE



Potential control of *Vespula germanica* and *Polistes dominula* (Hymenoptera: Vespidae) using entomopathogens

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ABSTRACT

The first recordings in South Africa of the invasive social wasps, *Vespula germanica* and *Polistes dominula* (Hymenoptera: Vespidae) were in 1974 and again in 2008. These wasps are known to represent a significant threat to the biodiversity of ecosystems and socio-economic activities in countries they invade. Here, the susceptibility of *V. germanica* and *P. dominula* larvae to indigenous species of entomopathogenic nematodes (EPNs), namely *Heterorhabditis bacteriophora*, *H. noenieputensis* and *Steinernema yirgalemense*, and entomopathogenic fungus (EPF) *Beauveria bassiana* was tested both in laboratory bioassays and *in situ* wasp nests application in the Western Cape province. The bioassay results indicate that both *P. dominula* and *V. germanica* larvae are susceptible to all biocontrol agents used. All larvae of both wasp species were dead and infected after exposure to 200 IJ/insect within four days after inoculation. Similar results were obtained seven days after inoculation with *B. bassiana*. To test in-field pathogenicity against *P. dominula*, wasp nests were treated with *B. bassiana*, *H. bacteriophora*, a mixture of the EPF and EPN species and a control of distilled water. The number of larvae and pupae infected 168 h after application were determined. In both cases, the mixture of EPF/EPN performed best, with a mean percentage of $31.39\% \pm 4.75\%$ larvae infected, and only $3.42\% \pm 0.68\%$ of pupae infected. The results obtained suggest that higher concentrations of selected biologicals can be used as inundative biological control agents within an integrated management programme for the control of *P. dominula* and *V. germanica* in South Africa.

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Biological control; EPF; EPN; wasp nests; laboratory screening; invasive wasps

Highlights

- At 48 h after inoculation, 100% of the wasp larvae treated with EPN were infected
- After 168 h, 74–98% of EPF-treated larvae showed visual infection of *Beauveria bassiana*
- *Heterorhabditis bacteriophora* outperformed other biocontrol agents tested
- Combined EPN and EPF *in situ* nest application achieved larval infection of $31\% \pm 4.75\%$.

Introduction

Since the first recordings in South Africa of the highly invasive wasp species, the European/German wasp *Vespula germanica* (Fabricius) in 1974 and the European paper wasp *Polistes dominula* (Christ) (Hymenoptera: Vespidae) in 2008, little research until recently has been done on these wasp species (De Villiers et al., 2017; Haupt et al., 2019; Roets et al., 2019; Tribe & Richardson, 1994; Van Zyl, 2016; Van Zyl et al., 2018; Veldtman et al., 2021). According to the 'National Environmental Management: Biodiversity Act' (NEMBA, 2004), *V. germanica* is categorised as a 1b invasive species in South Africa, which means that an invasive species management plan must be in place. *Vespula germanica* is an active predator and could pose a significant ecological threat especially in invaded environments (Tribe & Richardson, 1994). *Polistes dominula* is known to outperform native congener species in South Africa (Benadé, 2016), they have a propensity to use man-made structures as nest-building sites which makes them a general nuisance (Donovan, 1992) and with this species being a relatively new arrival in South Africa, its ecological impact is unknown. The still limited distribution range of the two wasps in South Africa (Benadé et al., 2014; Van Zyl et al., 2018; Veldtman et al., 2021) creates a favourable scenario for management efforts if they are acted on rapidly, and it increases the possibility of successful control to limit further spread (Van Zyl et al., 2018).

Various control methods have been developed and implemented internationally to curb the population expansion of mostly *V. germanica* and *Vespula vulgaris* (Linnaeus) (Hymenoptera: Vespidae), and, to a lesser extent, other social wasps (Glare et al., 1996; Moller et al., 1991a, 1991b). Chemical control of nests and adults has been proven to achieve only temporary abatement within a localised area (Duthie & Lester, 2013). Even though the chemical control of wasps has been successful in some instances, its use is accompanied by potential disadvantages such as the human and pets risk of application in urban areas (Merino et al., 2007; Rose et al., 1999; Van Zyl et al., 2018). Similarly, several attempts at controlling invasive wasps using parasitoids have not been successful, which can be ascribed to the wasps being eusocial, with a high reproduction capacity and specialised behaviour such as colony cleaning (Beggs et al., 2011; Donovan et al., 2002). It is, therefore, of importance that the available and new control options be improved and expanded on.

Compared to chemical control, the biological control of wasps is an understudied area. Significant biological control of invasive ants, termites and wasps has been achieved (Glare et al., 1996; Mayorga-Ch et al., 2021; Milner & Staples, 1996; Rose et al., 1999). Although the attempts to control social wasps using biocontrol agents have, in many cases, not been successful in reducing the abundance of populations, because of social behaviour and their resilience (Donovan et al., 2002; Martin, 2004). In other instances, it has been stated that techniques (toxic baits, synthetic lures, parasitoids) followed could be improved (Lester et al., 2013) to enable the true reflection of the efficacy of the agent. Occasionally, several follow up releases of a parasitoid are required to obtain successful control (Beggs et al., 2008).

Entomopathogenic fungi (EPF) that have specifically demonstrated potential against social wasps, such as *V. vulgaris*, *V. germanica* and *Polistes hebraeus* Fabricius, include *Aspergillus flavus* (Link), *Beauveria bassiana* (Balsamo-Crivelli) as well as *Metarhizium anisopliae sensu lato* (Metchnikoff) (Hypocreales: Clavicipitaceae) isolates (Harris

et al., 2000; Jaronski, 2010; Lacey et al., 2001; Merino et al., 2007; Reason et al., 2022). Although *B. bassiana* is the most widely distributed species of its genus and has an extensive host range being prevalent in many arthropods, isolates of the same species have a restricted host range (Goettel et al., 1990; Zimmermann, 2007). Scant attention has, however, been paid to the pathogenicity to *P. dominula* (Paynter & Ward, 2012) of *B. bassiana*. Moreover, there is still a lack of thorough assessments of the likely impact that the specific isolates might have on non-target hosts (Brownbridge et al., 2009; Harris et al., 2000; Mayorga-Ch et al., 2021; Merino et al., 2007).

Specific EPN species that have been confirmed to be pathogenic against Vespidae are *Steinernema carpocapsae* Wouts (Glare et al., 1993, 1996), *Steinernema feltiae* Wouts and mermithid nematodes, like *Pheromermis pachysoma* (Von Linstow) and *Pheromermis vesparum* (Kaiser) (Rose et al., 1999). Preliminary research showed *Steinernema carpocapsae* (Weiser), *S. feltiae* and *Heterorhabditis bacteriophora* Poinar can cause over 70% mortality of *Vespula pensylvanica* under laboratory conditions (Gambino et al., 1992; Guzman, 1984). Some species of *Steinernema* and *Pheromermis* have been isolated from *Vespula*, males, drones and queens, and confirmed as parasites of Vespidae (Glare et al., 1993, 1996; Martin, 2004; Moller et al., 1991a, 1991b; Rose et al., 1999).

The first objective of the current study was to determine, by means of laboratory bioassays, the pathogenicity of three local EPN and one commercially produced EPF against *V. germanica* and *P. dominula* larvae in South Africa, and to assess the potential use of the agents involved as inundative biocontrol agents. The second objective of this study was to determine the in-field virulence of the biological control agents that have proven to be pathogenic to larvae of *P. dominula* in laboratory conditions. This is a first step to developing a biocontrol programme against invasive wasps in South Africa.

Materials and methods

Source of insect hosts and EPNs

Galleria mellonella (Linnaeus) (Lepidoptera: Pyralidae), wax moth larvae (WML) were used as hosts for the *in vivo* production of infective juveniles (IJs) of EPN. Insect larvae were reared on a wheat-based diet in ventilated plastic containers (31.5 × 20.5 × 7 cm) at 25°C in the dark (Van Zyl & Malan, 2015). The EPN species used in the bioassay trials and field trials were all locally isolated EPN species obtained from the collection at the Department of Conservation Ecology and Entomology, Stellenbosch University (Table 1). The nematodes were cultured at 25°C according to the method adopted by Kaya and Stock (1997). The IJs were harvested from infected WML using White traps (Gaugler et al., 2000) and were then stored in 150 ml distilled water in vented 500-ml culture flasks (Nunc™ Cat. No. 144530) at 14°C in the dark, aerated on a weekly basis, and used within a month after harvesting. For the bioassay trials, nematodes were applied at a concentration of 200 IJs per insect larva. For the field trial, nematodes were concentrated (300 IJs/50 µl) the day before field application, using the method of Glazer and Lewis (2000).

Source of EPF

The *B. bassiana* fungus strain (R444) used in the trials is a commercially available product obtained from Andermatt Plant Health Products (Pty) Ltd., called Eco-Bb.

Table 1. Entomopathogenic nematodes and the entomopathogenic fungus used in the trials, with their isolate number, GenBank accession number, original sample locality and habitat.

EPN and EPF species	Isolate	GenBank accession number	Origin	Habitat
<i>Beauveria bassiana</i>	R444	N/A	Clanwilliam, Western Cape	N/A
<i>Heterorhabditis bacteriophora</i>	SF351	FJ455843	Wellington, Western Cape	Vineyard
<i>H. noenieputensis</i>	SF669	JN620538	Noenieput, Northern Cape	Soil under fig tree
<i>Steinernema jirgalemense</i>	157-C	EU625295	Nelspruit, Mpumalanga	Citrus orchard

The product is registered for use on whitefly, including the woolly whitefly (*Aleurothrixus floccosus*), red spider mites (*Tetranychus urticae*), tomato tuber moth (*Phthorimaea operculella*), false codling moths (*Thaumatotibia leucotreta*) and mealybugs (*Pseudococcidae* spp.). Prior to use, it needed to be stored in a refrigerator at 7°C. For the bioassay trial, the product was mixed with distilled water.

Source of wasps for the bioassay trials

The larvae of both *P. dominula* and *V. germanica* were collected from live nests in the field in Stellenbosch, South Africa (33.9200°S, 18.8600°E), during the spring and summer (October to February 2016), and autumn (March). Protective suits were worn against the unsubdued wasps, which were present in and around the nests, while removing the nests. In the case of the *P. dominula* nests, the whole nest (and, in some instances, more than one nest) was removed by hand, taking care neither to damage the nest cells or to touch the larvae present. In all cases, a beehive tool was used to dislodge the petiole. The *V. germanica* nests were generally larger than the *P. dominula* nests, and therefore, only a few combs of a nest were used at any one time, that contained enough larvae to conduct a trial. In the case of subterranean *V. germanica* nests, the whole nest was excavated, with all the comb layers being collected. All live adult wasps were removed from the nests before being placed in aerated plastic containers that were transported back to the laboratory. Larval material from both the aerial and the subterranean nests was used in the case of *V. germanica*, with the required material being taken from aerial nests in the case of *P. dominula*. The wasp larvae were removed from the nest cells at room temperature (25°C), using forceps, and placed in 24-well (Greiner bio-one, Cellstar®, Cat. No. 662160), flat-bottom bioassay plate wells, after which they were treated immediately.

Bioassay protocol

Four treatments were applied, consisting of three EPN treatments and one EPF treatment. A total of ten 24-well bioassay plates were used per treatment, of which five were controls, in which the larvae were treated with water only. Ten alternately spaced wells of each 24-well bioassay plate were each lined with three disks of filter paper (Whatman No. 1), with one late instar uncapped wasp larva being added per well after the EPN, EPF or water only suspensions were respectively applied directly to the filter paper ($n = 50$ insects/treatment). The fungal inoculum was prepared according to the instructions on the product label. Fungal spores were applied with an Eppendorf micropipette, at 1 g (2×10^9 spores) of the commercial product per 1 L of distilled water in 50 µl, whereas the EPN were applied at a concentration of 200 IJs per insect larva.

For the field trial, the dosage was doubled to 2 g of the commercial product per 1 L of distilled water. The specific concentrations were selected, due to the optimal dosage of the EPF and EPN isolates used against the wasps not being known. The IJ concentration involved has been used as the base dosage in several different bioassay trials against other insects (Navon & Ascher, 2000). Only the larval stage of the wasps was treated with biocontrol agents which is a more practical control measure to adopt (Gentz, 2009).

The bioassay plates were then incubated in closed plastic containers (22 × 15.5 × 8 cm) in the dark at 25°C and at relative humidity (RH) of > 80%. After 24 and 48 h the number of dead wasp larvae were recorded for each treatment, by means of gently probing the larvae with forceps. No movement in response to stimulation confirmed mortality. After 48 h, the filter paper disks in each well were replaced with fresh, moist disks and incubated for a further 48 h in the case of EPN-treated larvae. Mortality due to infection was confirmed after 96 h by means of dissection, and nematode visually observed inside of the larvae, with the aid of a dissecting microscope. For EPF-treated larvae, incubation occurred for a further 120 h and mortality due to fungal infection was confirmed after 168 h (7 days), through the visual inspection of mycosed cadavers under a dissection microscope, and through visual observation of the cadaver colour change to a pale pink. The trial was repeated on a different test date with fresh batches of entomopathogens, to ensure reproducibility of the results obtained (Butt & Goettel, 2000).

Application protocol in situ nests

Polistes dominula nests ($n = 96$), taken from sites in Stellenbosch and Elsenburg, were selected for use in the nest application. Temperature and humidity iButton® sensors (Hygrochron®, Fairbridge Technologies) were placed next to the nests, to log ambient temperature and relative humidity (RH) every minute for 24 h, from 07h00 on the first morning, when the nests were treated, to 07h00 the following morning. Medium-sized nests, with an average of 361 cells and mostly situated below roof eaves, were first sprayed with distilled water, before being treated with the selected biocontrol agents at a concentration of 300 IJs/50 µl and fungi at 2 g commercial product per litre of water. The trial took place during late summer and autumn (February to April), which is a period when *P. dominula* wasps are abundant and their nests tend to be well established. An average of 7.75% of the nests consisted of larvae, with 11.08%, on average, containing pupae. To confirm viability, treatment samples were sprayed into vials using a hand-held spray applicator, before being sprayed onto nests. With the aid of a dissecting microscope, the IJs that moved either voluntarily, or in response to prompting by means of forceps, were classified as viable specimens. The presence of fungal spores in spray samples was assumed viable, since it was obtained from a commercially formulated product. Four different treatments were applied to the respective nests: (1) a control treatment of distilled water only; (2) a *H. bacteriophora* suspension, at a concentration of 300 IJs/50 µl; (3) a *B. bassiana* suspension of 2 g commercial product/1 L water; and (4) a mixture of *H. bacteriophora* and *B. bassiana*, at the same concentrations as the separate treatments. In the bioassay tests conducted beforehand to determine the pathogenicity of the specific biocontrol agents against *P. dominula* larvae, lower concentrations of *B. bassiana* (1 g commercial product/1 L water) and *H. bacteriophora* (200 IJs/50 µl) were used. Increased concentrations for

field applications were used to compensate for biocontrol agent loss in the field when using spraying equipment.

The respective nests were treated with the specific agent and sprayed on the outside until all the cells were filled with the suspension until visual drip-point. Nests sprayed with different treatments were not placed adjacent to each other, to avoid cross-contamination. After application of various treatments, nests were kept moist for a period of 6 h, by spraying it with distilled water every 2 h. After 24 h, the nests were collected in separate, moist paper towel-lined plastic containers (22 × 15.5 × 8 cm), placed on overturned Petri dishes (90 mm diam.), to avoid contact between the paper towelling and the nest surface. Nests were then returned to the laboratory, where they were incubated at 25°C, with a day/night regime of 14:10, after the number of cells, larvae and capped cells (pupae) were counted.

Daily readings of the number of emerging wasps, of the newly formed wasp pupae and of the remaining wasp larvae were taken up to 144 h (6 days) after treatment application. Wasps were continuously removed from the containers as they emerged from the capped cells, with them then being placed in the alternate filter paper-lined wells of the 24-well bioassay plates and incubated at 25°C for an additional 24 h, to allow for the potential development of fungal growth or nematode infection. After 144 h, all the larvae and pupae treated with *B. bassiana* were removed from the nest cells, using forceps, and then placed into filter paper-lined bioassay wells, and incubated for an additional 24 h at 25°C, where they were observed for the development of fungal growth.

Data analyses

A repeated-measures ANOVA was used to compare the mortality (after 24 and 48 h) and infection levels (after 96/168 h), respectively over time, to assess differences in the effect of the various EPN and EPF species on both species of insect larvae. Additionally, significant differences occurring between Trial 1 and 2 could be compared after performing the Repeated Measures of Analysis of Variance (ANOVA). An arcsine transformation was done on the square root of the proportion of dead and infected larvae at each timeframe, to stabilise their variances and subsequently bringing them closer to normality. The Shapiro–Wilk test was used to test the normality of the data. Significant differences between treatment means were determined by means of Fisher's Least Significant Different (LSD) method. For the *in situ* nest application results a Repeated Measures ANOVA, with a LSD test, were used to compare the mean percentage infection of *P. dominula* larvae and pupae, 168 h after application of the respective treatments. A post-hoc comparison of means, using the bootstrap multi-comparison test, was then performed (Efron & Tibshirani, 1998). All the statistical analyses were performed using STATISTICA, version 12 (StatSoft Inc., 2013). Mortality of insects equals infection, which was visually determined.

Results

Bioassay trials

Pathogenicity of entomopathogens against *V. germanica* larvae

As a significant difference ($F_{(1, 70)} = 6.99$, $p = 0.01$) was observed between Trial 1 and Trial 2 virulence trials in the main effects of time and treatment, the results of Trials 1

and 2 could not be pooled and were analysed separately. In Trial 1, 24 h after inoculation with EPNs, the proportion of dead larvae that had been treated with *S. yirgalemense* was the highest, compared to the other treatments, with the proportion being the lowest for the *B. bassiana*-treated larvae (Figure 1(A)). Significant differences were also found between the proportion of dead larvae after 24 h, as treated with *S. yirgalemense*, *H. bacteriophora* and *H. noenieputensis*, respectively. At 48 h after inoculation, 100% of the EPN-treated larvae were dead. Infection of all the EPN-treated larvae with the

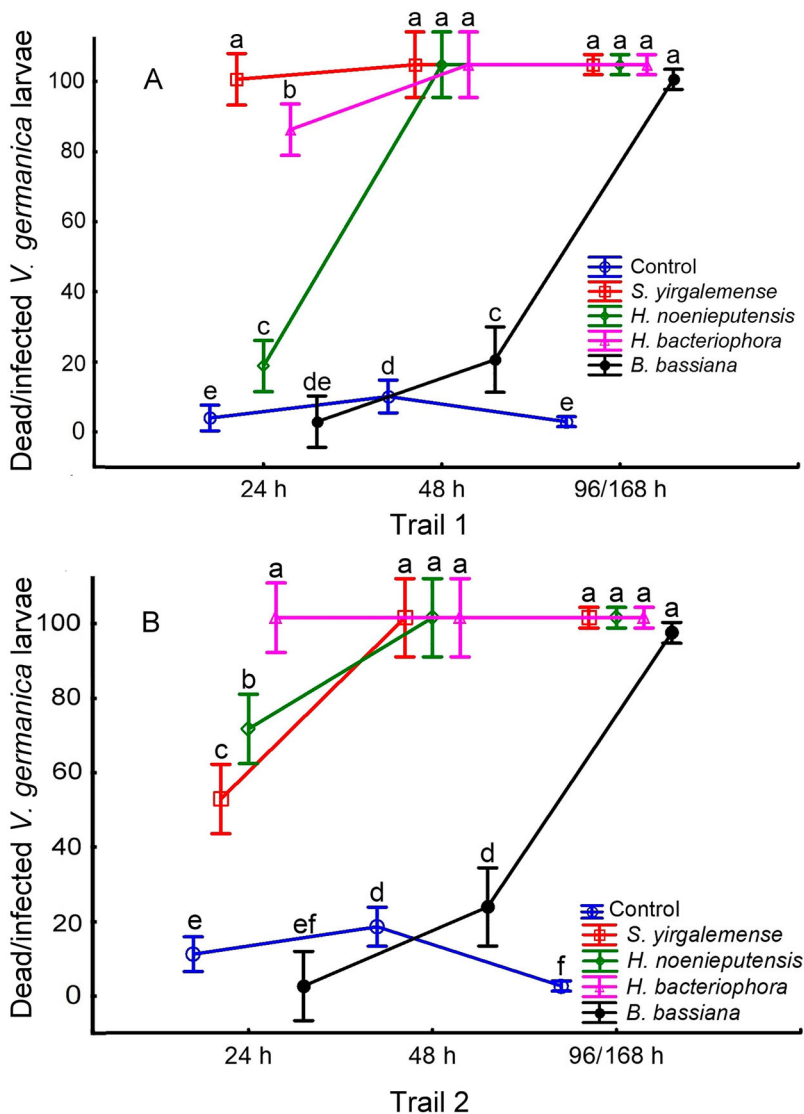


Figure 1. The mean number (95% confidence intervals) of dead (24 and 48 h) and infected (96/168 h) *Vespula germanica* larvae treated in laboratory bioassays with *Steinernema yirgalemense*, *Heterorhabditis noenieputensis*, *H. bacteriophora*, *Beauveria bassiana* and control (repeated measures ANOVA; (A) Trial 1: $F(8, 70) = 83.188$, $p < 0.01$; (B) Trial 2: $F(8, 70) = 60.761$, $p < 0.01$). The same letters above the bars indicate no significant difference ($p > 0.05$) between time and dead or infected insect larvae.

three EPN species was confirmed 96 h after inoculation. After 168 h, $98\% \pm 0.7\%$ of the EPF-treated larvae had been infected with *B. bassiana*. The percentage dead and infected larvae in the corresponding controls of all biocontrol treatments were below 10.5% for both trials.

In Trial 2, unlike in the case of Trial 1, the larvae treated with *H. bacteriophora* showed the highest mortality after 24 h, whereas the larvae treated with *S. yirgalemense* showed the second-lowest mortality (Figure 1(B)). The lowest proportion of dead larvae after 24 h was found for the *B. bassiana* treatment. A similar trend to Trial 1 was observed in Trial 2, 48 h after inoculation, with all the EPN-treated larvae being dead. All EPN-treated larvae were also infected 96 h after inoculation.

The same number of larvae treated with the EPF in Trial 1 ($98.00\% \pm 0.7\%$) were infected with the fungus 168 h after inoculation. The two graphs (Figure 1) show the difference in the level of mortality caused by the three different EPN species at 24 h after inoculation in Trial 1, compared to Trial 2. However, after 48 h the EPF showed no difference between the two trials with lower mortality than EPN.

Pathogenicity of entomopathogens against *P. dominula* larvae

Significant differences were observed in the main effects of time and treatment between the first and second trial ($F_{(1, 70)} = 19.111$, $p < 0.001$), therefore the results for Trials 1 and 2 cannot be pooled and were analysed separately. In Trial 1, the treatment that caused the highest larval mortality after 24 h was *H. bacteriophora* (Figure 2(A)). *Heterorhabditis noenieputensis* and *B. bassiana* had the lowest larval mortality 24 h after inoculation. The three EPN treatments killed 100% of larvae 48 h after inoculation, and infection by dissection was visually confirmed after 96 h. Larvae treated with *B. bassiana* achieved a significantly lower larval infection ($74\% \pm 2.3\%$) as taken at 168 h after inoculation.

In Trial 2 (Figure 2(B)), a similar trend in terms of highest larval mortality and infection achieved by each treatment through time, was observed when compared to Trial 1. At 24 h after inoculation *H. bacteriophora* killed the highest number of larvae, followed by *S. yirgalemense* and *H. noenieputensis*. All EPN-treated larvae died after 48 h, while only $44\% \pm 6.1\%$ larvae treated with *B. bassiana* had died after the same time. At the last recording measuring the number of larvae infected by EPN and the EPF at 96 and 168 h, respectively, 100% of EPN-treated larvae were infected with the respective EPN species, while the EPF treatment caused $82\% \pm 2.7\%$ larval mortality due to fungal infection. The percentage larval mortality and infection obtained in the corresponding controls for each biocontrol treatment, was below 15.5% over time for both trials.

In situ trial

Susceptibility of larvae in situ nests

The RH in the field for all sites ranged between 46% and 73% during the time from application until the nests were removed 24 h later. Although the average temperature was between 18°C and 31°C during the trial period, a high of 58°C was once logged on a day in late February. To keep the nests moist, they were sprayed with distilled water before the treatments were applied, as well as up to 6 h after application. Results from the complete nest treatments against larvae of *P. dominula* in situ differ significantly

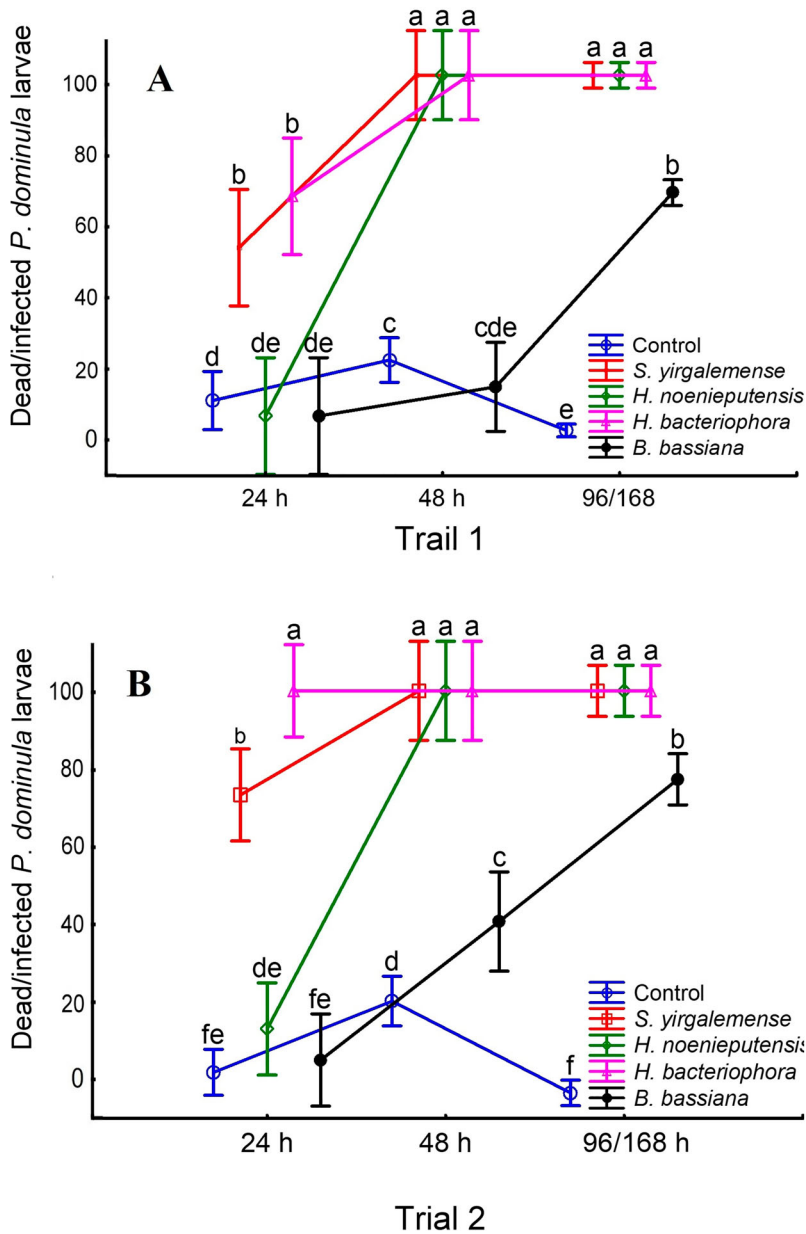


Figure 2. The mean number of dead (95% confidence level) (24 and 48 h) and infected (96/168 h) *Polistes dominula* larvae treated in laboratory bioassays with *Steinernema yirgalemense*, *Heterorhabditis noenieputensis*, *H. bacteriophora* (200 infective juveniles/insect), *Beauveria bassiana* (2×10^9 spores/g) (repeated measures ANOVA; (A) Trial 1: $F(8, 70) = 30.693$, $p < 0.001$; (B) Trial 2: $F(8, 70) = 24.976$, $p < 0.001$). The same letters above the bars indicate no significant difference ($p > 0.05$) between the time and the dead and infected insect larvae.

from each other ($F_{(3, 96)} = 9.256$, $p < 0.001$). Nests that were treated with a mixture of *H. bacteriophora* (EPN) and *B. bassiana* (EPF), showed the highest level of larval infection ($31.39\% \pm 4.75\%$), with the level differing significantly from nests treated with the

control ($p < 0.001$), whose level also, in turn, differed significantly from that of all other treatments tested (Figure 3). Even though the percentage of larval infection obtained after treatment with *B. bassiana* was slightly higher than the infection levels of *H. bacteriophora*, the treatment results were not found to differ significantly from each other ($p = 0.429$).

Susceptibility of the pupae in situ nests

In all the treatments, the percentage of *P. dominula* pupae infected with biocontrol agents was much lower compared to the percentage of larvae that were infected in the nests. A similar trend, compared to that of larval infection, was observed in terms of the efficacy of each of the treatments. The highest mean percentage ($3.43\% \pm 0.68\%$) of infected pupae was obtained in the nests that were treated with a combination of EPN and EPF. Significant differences were found between the treatments and the percentage of pupal infection obtained ($F_{(3, 93)} = 5.001$, $p = 0.003$). Only the joint treatment of EPN and EPF differed significantly from the control treatment ($p = 0.009$). No significant difference was found between the other treatments.

Larval development into pupae and into adult wasps from treatment nests

No significant difference was observed in the percentage of *P. dominula* larvae that developed into pupae after the various treatments were applied ($F_{(3, 93)} = 0.097$, $p = 0.961$). A mean of $25.38\% \pm 4.59\%$, $26.34\% \pm 4.59\%$, $28.77\% \pm 4.78\%$, $27.48\% \pm 5.11\%$ larvae developed to pupae after nests were treated respectively with distilled water, *H. bacteriophora*, *B. bassiana*, and a combination of the EPF and EPN. Similarly, no significant differences were observed in the percentage pupae that developed to adult wasps after the various treatments were applied ($F_{(3, 93)} = 0.86157$, $p = 0.46400$). A mean of $29.03\% \pm 24.38\%$; $23.30\% \pm 17.32\%$; $34.85\% \pm 35.02\%$; $27.39 \pm 23.16\%$ pupae developed to adults after

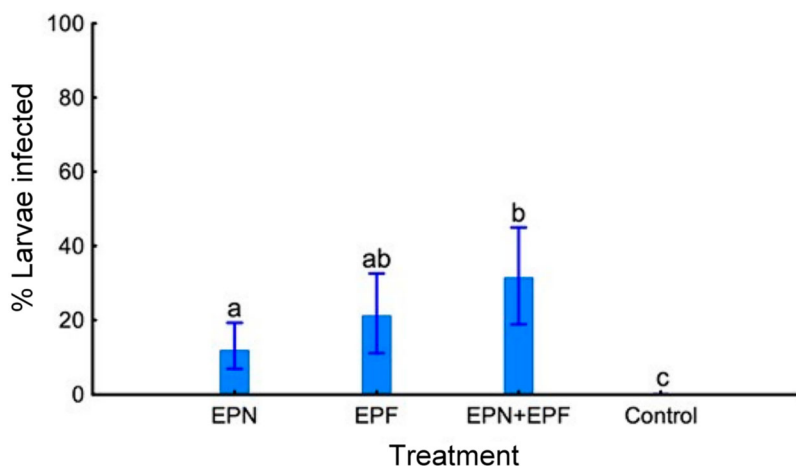


Figure 3. Mean percentage (95% confidence level) infected *Polistes dominula* larvae, recorded 168 h after nests being treated *in situ* with EPN, EPF and a combination of EPN and EPF (one-way ANOVA; $F_{(3, 96)} = 9.256$, $p < 0.001$). The same letters above the bars indicate the presence of no significant difference ($p > 0.05$) between the treatment and the percentage of infected insect larvae in the nests.

nests were treated respectively with distilled water, *H. bacteriophora*, *B. bassiana* and a combination of the EPN and EPF.

Discussion

Significant differences were found among each of the treatments that were applied in both trials. The noteworthy difference was the rate at which mortality occurred. When comparing the different EPN treatments, a faster mortality rate can be seen as an indication of a higher level of host/pathogen pathogenicity (Brodeur, 2012). *Heterorhabditis bacteriophora* was found to have caused the highest larval mortality after 24 h exposure over both trials on average, compared to all the other agents tested, and was therefore, the preferred agent selected for application in the follow-up field trial of complete nest treatment.

As expected, compared to the EPN treatments, *B. bassiana*, performed the slowest in terms of killing the larvae over time, due to its different mode of infection. Apart from host susceptibility, the time it took to kill can also be related to the concentration involved (Butt & Goettel, 2000). The filter paper method of inoculation used in the laboratory can serve as a form of standardisation, however, much higher fungal concentration should be considered for *in situ* nest application. In previous studies, higher spore concentrations were found to lead to raised levels of susceptibility (Harris et al., 2000).

At the last observation, 168 h after inoculation, the fungus showed overt mycosis on the host cadaver, but the levels of *V. germanica* larval infection in both trials were lower compared to those found with the larvae treated with EPN. The *B. bassiana* strain used in the current study remains an attractive control option, with, in some cases, its delayed action possibly being seen as a positive attribute. This is especially relevant in the case where the success of wasp control partly relies on the adult wasp remaining uninhibited by the mycopesticide until it reaches the colony, thereby inducing horizontal transmittance throughout the nest (Harris et al., 2000; Merino et al., 2007).

Even though all three EPN caused mortality and infection of all the *P. dominula* larvae, significant differences were found to have occurred between treatments at the specific timings, with *H. bacteriophora* causing the highest mortality at 24 h after inoculation. The evidence that *H. bacteriophora* repeatedly outperformed the other biocontrol agents might indicate a good host–pathogen match. In contrast, *H. noenieputensis* was found to be the least pathogenic of all the EPN agents. This level of mortality was comparable to that caused by the fungus at the same stage.

A similar pattern to what was observed with the *V. germanica* larvae inoculated with *B. bassiana* was reflected in the trials for the *P. dominula* larvae. Even though the level of infection achieved after 7 days was high at the last recording, it was lower than the level of infection obtained in *V. germanica* larvae, which can potentially be attributed to fungal toxins being less pathogenic (Harris et al., 2000; Merino et al., 2007). The conclusion can therefore be drawn that this specific strain of *B. bassiana* showed a higher level of virulence towards *V. germanica* larvae than it did towards *P. dominula* larvae.

In comparison, the results obtained from complete nest application showed the mean percentage of infected *P. dominula* larvae to be much lower than it was for both EPF and the EPN treatments, respectively, compared to the results that were obtained under

laboratory conditions. Only a maximum of 31% of larval infection was obtained when a combination of the EPN and EPF species were applied to nests in the field. While the infection rate of the larvae from the combined treated nests was low, it was observed that the fungus infected a greater number of larvae (77%) compared to the nematodes (23%). This difference in infection rates could be attributed to the fungus's potential resilience to the field conditions, surpassing that of the nematodes.

High humidity levels (>70% RH) are required for at least 8–24 h after the application of such agents as EPNs and EPF, to ensure successful germination and infection (Arthurs et al., 2004; Georgis & Gaugler, 1991; Zimmermann, 2007). As humidity levels were raised artificially above natural conditions it could have improved infection levels. Regardless of the high temperature, the EPF and EPN were still capable of infecting the larvae. The optimal temperature for *H. bacteriophora* to infect insect hosts is $\pm 25^{\circ}\text{C}$, with the EPN rendered inactive or die when exposed to some very high temperatures experienced during the field trial (Grewal et al., 1994). The range of temperature for survival of *B. bassiana* is between -30°C and 20°C (Roberts & Campbell, 1977; Tefera & Pringle, 2003), while at temperatures higher than 50°C , the fungus is killed (Walstad et al., 1970). To minimise the influence of high temperatures on biocontrol agent efficacy, the agents were applied early in the morning. In future studies, sub-optimal in-field conditions could be overcome using adjuvants such as surfactants or incorporating humectants in product formulations.

Biotic constraints, like the behavioural characteristics of social wasps, further tend to impede the ability of biocontrol agents to infect progeny. Wasps actively suppress growth and dispersal of biocontrol agents in their nests, by removing diseased nest mates, and, in addition, the presence of antimicrobial compounds in wasp saliva, which is embedded in the cell walls of nests, inhibit fungal development (Rose et al., 1999). With *P. dominula* nests not being enclosed in a papery layer, such as in the case of *V. germanica* nests, the removal of infected material is facilitated which in turn lowers the persistence of biocontrol agents in nests. Hence, it is important to consider that in this study, the extraction of adult wasps from incubated nests could have been either favourable or unfavourable.

Despite levels of larval infection obtained being low after field application, significant differences in the pathogenicity of the different treatments were noticeable. The treatment consisting of both EPN and EPF performed best, with it being the only treatment to differ significantly from the rest. Several studies have reported observing a synergistic effect when EPN, EPF and chemical pesticides are used in combination with each other to combat insect pests (Ansari et al., 2010; Barbercheck & Kaya, 1990; Ericsson et al., 2007). In a study conducted by Shairra and Noah (2014), treating the cotton leaf worm (*Spodoptera littoralis*) (Lepidoptera: Noctuidae) with combinations of EPF species (*B. bassiana* and *Metarhizium anisopliae*) and EPN species (*H. bacteriophora* and *S. riobrave*), increased the mortality levels in most combinations compared to when each pathogen was applied individually.

The results obtained from this study clearly show that the highest pathogenicity of the selected biocontrol agents is likely to be achieved when applying it to *in situ* nests of which the cells contain mainly larvae (the beginning of the wasp season in Spring), since larvae were overall more susceptible to the biocontrol agents tested than the pupae. With field applications, low infection levels of both larvae and pupae were obtained when treating *P. dominula* nests with indigenous *H. bacteriophora* and *B. bassiana* species. Therefore,

several aspects of the infield use of such EPN and EPF agents should be improved to increase their efficacy. Proteins that are present in wasp saliva contribute to the hydrophobic-produced paper of which the wasp nests are built (Espelie & Himmelsbach, 1990) and could to some extent, be responsible for poor adhesion of biocontrol agents to nests, especially when the agents are applied in aqueous form. To increase the extent of coverage and limit desiccation, adjuvants can be incorporated into biocontrol formulations, to increase levels of humidity and reduce sedimentation of organisms in aqueous solutions (De Waal et al., 2013; Van Niekerk & Malan, 2015).

Invasions should be assessed on a case-by-case basis when the goal is to develop a tailored management programme. In this case, the use of indigenous EPF and EPN for wasp control should be considered, where other species and strains are screened in the laboratory, to obtain the most effective candidates prior to the commencement of field trials. Since a very small amount of biological agent should prove to be sufficient for nest treatment, the concentration, especially in the case of EPNs, could be increased substantially without exposing the urban environment to risk. The confirmed laboratory effectiveness of the pathogenic control agents provides the basis for future research to identify effective methods of application of the pathogens in the field. The application of pathogens through non-lethal traps is another avenue of research that could improve the practicality of using biological control for limiting invasive social wasp species.

This research contributes to the understanding of using indigenous biological control agents to manage two invasive wasp species in South Africa. It also emphasises the need for careful evaluation and adaptation of control methods to suit the ecological and environmental conditions of a specific region affected by invasive wasps. Achieving high wasp pathogenicity under field conditions should serve as a stepping-stone towards optimising the control programmes of invasive wasps in South Africa and their treatments.

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author, [CvZ].

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